Supplementary Information for "Resistance-Aware Computational Analysis of Antimalarial Leads from African Natural Products"

August 10, 2026

Table 1: Docking protocol validation: enrichment metrics and re-docking accuracy. Values reproduce the corrected benchmarks from the parent study (V2-corrected grids; AutoDock Vina + DiffDock consensus). MMV Malaria Box values are consensus-score ROC-AUCs for the three targets with MMV docking data (PfDHFR, PfCRT, PfATP4); DEKOIS 2.0 validation used AutoDock Vina only for PfDHFR. BEDROC values use $\alpha = 20$ (MMV). Re-docking RMSD is the heavy-atom RMSD between the top-scoring docked pose and the co-crystallised ligand; MTX redocking failed (RMSD $\approx$ 30 Å).

Target	Method	ROC-AUC	EF1 %	EF5 %	BEDR
PfDHFR (7F3Y)	Vina redock	–	–	–	–
	MMV consensus (Vina+DiffDock)	0.924	–	2.57	1.30
	DEKOIS (Vina only)	0.450	–	0.00	–
PfCRT (6UKJ)	MMV consensus (Vina+DiffDock)	0.971	–	1.11	9.43
PfATP4 (9N10)	MMV consensus (Vina+DiffDock)	1.000	–	2.00	1.81
PfClpR (4GM2; not PfClpP)	MMV enrichment	not evaluated	–	–	–

Identity note.—PDB 4GM2 is PfClpR rather than PfClpP; the historical 4GM2 docking record is therefore not PfClpP validation.

Note.—Redocking against V2-corrected grids gave 100% success for the five alignable co-crystallised ligands. MTX (PfDHFR co-crystallised ligand) could not be redocked and is reported as N/A. AutoDock Vina + Gnina consensus scoring was used in the present validation study; the enrichment metrics above are taken from the parent study (Vina + DiffDock) to ensure consistency with the corrected P1 data.

Additional MD assessment. A diagnostic continuation of the parent PfDHFR and PfCRT systems did not yield a usable new trajectory. It therefore contributed no new MM-GBSA estimate or MD-RRS result. No candidate-specific Set-C production MD was obtained.

Use of Artificial Intelligence

During the preparation of this work the authors used AI assistants to support code development and data analysis scripting. The authors reviewed and edited all AI-assisted output and take full responsibility for the content of this article. No generative AI tool is listed as an author. All computational results, statistical analyses, and quantitative claims were generated and verified by the authors.

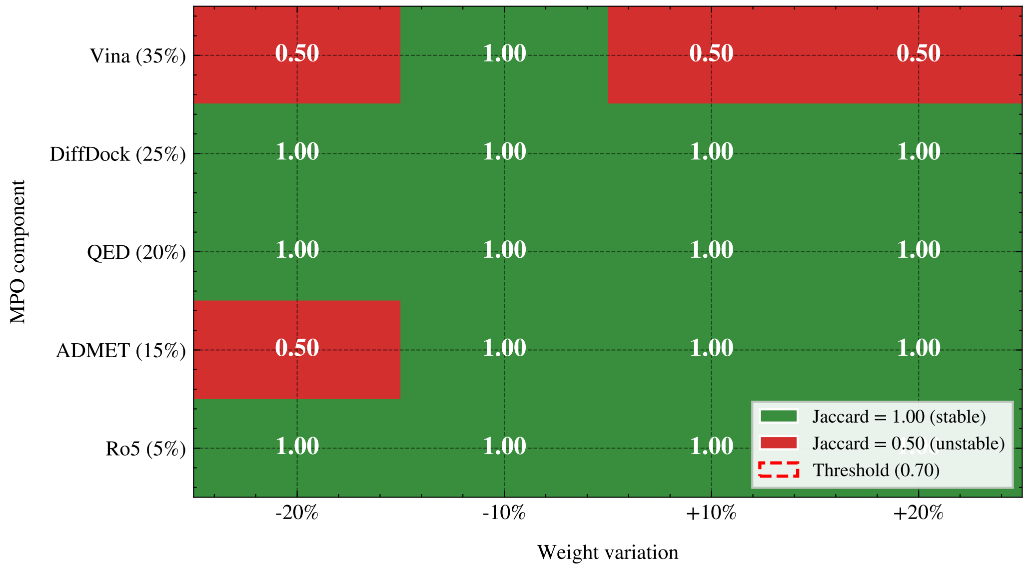

Figure 1: MPO sensitivity analysis heatmap. Each cell shows the Jaccard similarity of the top-20 hit list when the corresponding MPO weight is varied by ±20 %. Green (1.00) indicates the hit list is unchanged; red (0.50) indicates substantial reordering.

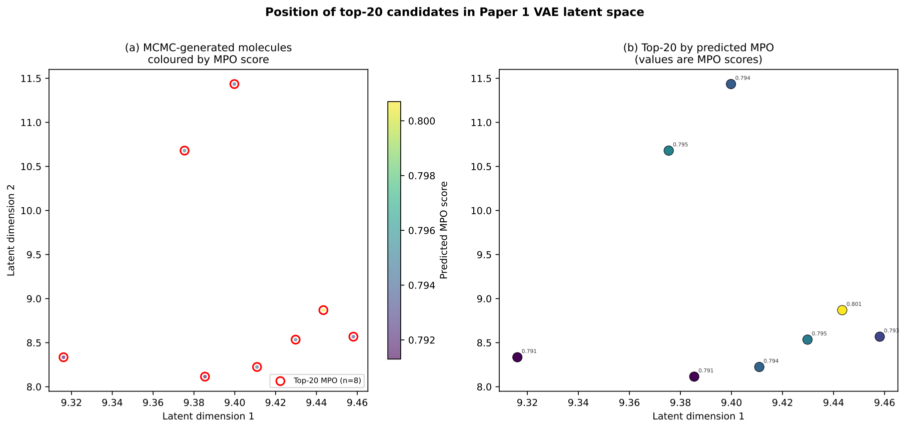

Figure 2: Position of the top-20 candidates in the Paper 1 VAE latent space. Molecules are coloured by predicted MPO score; top-20 by MPO are highlighted in red.

Table 2: Endpoint-analysis interpretation for the four parent-study consensus complexes used for production MD. These systems are distinct from the 17 set-C polypharmacology candidates. Mean ± standard deviation is shown only for the single interpretable MM-GBSA result from 101 snapshots; dissociated or conversion-corrupted systems are reported as N/A. Mutant RRS is a docking-based analysis of set C and is not assigned to these MD systems.

Ligand	Target	ΔG_{bind}	Std. dev.	Interpretation
164	PfClpR-labelled cohort*-WT	–	–	N/A; ligand dissociated‡
201	PfDHFR-WT	–	–	N/A; ligand dissociated‡
214	PfCRT-WT	−18.25	0.40	Interpretable (gmx_MMPBSA)
438	PfATP4-WT	–	–	N/A; conversion corrupted§

*PDB 4GM2 is PfClpR rather than PfClpP; the 164-labelled system is therefore not structural validation of

‡Ligand dissociated during production; no binding free energy is inferred.

§Excluded due to two-chain topology incompatibility with CHARMM36-to-AMBER conversion.

Table 3: Predicted ADMET profile of the 17 set-C polypharmacology candidates (ADMET-AI, parent-library screening). All values are machine-learning predictions; experimental ADMET profiling and three-tool concordance were outside the scope of this study. Range across candidates: LogS $[0.34, 0.73]$, CYP3A4 inhibition ≥ 0.5 for 7/17 candidates, Caco-2 permeability ≥ 0.5 for 15/17, Clint_h $[0.48, 0.63]$.

Compound	ADMET-AI LogS	CYP3A4 prob.	Caco-2 prob.	Clint_h
PP-01	0.34	0.39	0.63	0.60
PP-02	0.46	0.97	0.86	0.55
PP-03	0.43	0.62	0.95	0.55
PP-04	0.40	0.58	0.92	0.63
PP-05	0.61	1.00	0.35	0.51
PP-06	0.54	0.90	0.97	0.60
PP-07	0.36	0.02	0.75	0.62
PP-08	0.59	1.00	0.41	0.56
PP-09	0.65	0.13	0.81	0.57
PP-10	0.49	0.13	0.93	0.62
PP-11	0.39	0.06	0.96	0.58
PP-12	0.49	0.00	0.98	0.48
PP-13	0.57	0.97	0.98	0.53
PP-14	0.66	0.05	0.68	0.49
PP-15	0.66	0.49	0.95	0.51
PP-16	0.54	0.25	0.91	0.50
PP-17	0.73	0.01	0.82	0.50

Table 4: ACSI weight sensitivity analysis. Each of the four ACSI weights (w_{DrugBank}, w_{ANPDB}, w_{fsp^3}, w_{NPL}) was varied by $\pm 20\,\%$ while keeping the remaining weights proportional. The Jaccard index measures overlap of the resulting top-20 ACSI-ranked list with the baseline ranking. A Jaccard ≥ 0.70 (14/20 compounds stable) indicates robustness.

Weight varied	Variation (%)	Mean ACSI (top-20)	Std. dev.	Jaccard vs. baseline
Data to be populated after ACSI computation.				

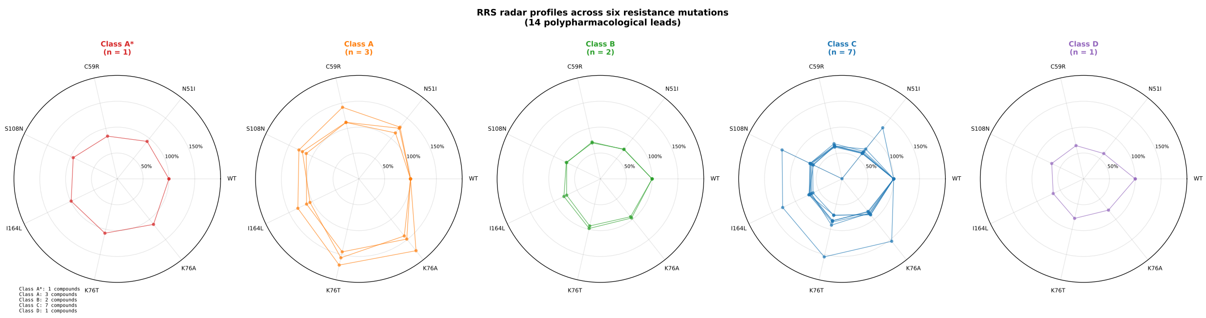

Figure 3: Per-compound, docking-derived RRS profiles across all six resistance mutations for the 17 set-C polypharmacological leads. The four parent-study MD systems are not included. Each axis represents one mutation; concentric rings indicate $50\,\%$, $100\,\%$, and $150\,\%$ RRS (WT-normalised). Class A* and Class A compounds fill the outer region (RRS $> 100\,\%$).